CAS SciFinder

Task History

Initiating Search

November 4, 2025, 2:43 PM

Search:

CAS Lexicon Search:

Horner-Wadsworth-Emmons reaction - Preferred Concept

Search Tasks

Task	Result Type	View
Returned Reference Results from CAS Lexicon (1,812)	References	View Results
Exported: Retrieved Related Reaction Results + Filters (1,310)	Reactions	View Results

Filtered By:

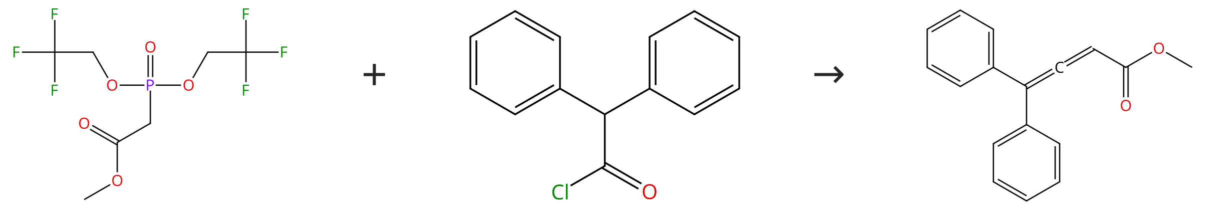
Yield:	90-100%, 80-89%, 70-79%
Document Type:	Journal
Language:	English
Publication Year:	2015 to 2025

CAS SciFinder

Reactions (50)

View in CAS SciFinder

Scheme 1 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (79) Suppliers (69)

31-339-CAS-14471581 Steps: **1** Yield: **100%**

1.1 **Reagents:** Bromo(1-methylethyl)magnesium
Solvents: Tetrahydrofuran; 1 h, 0 °C

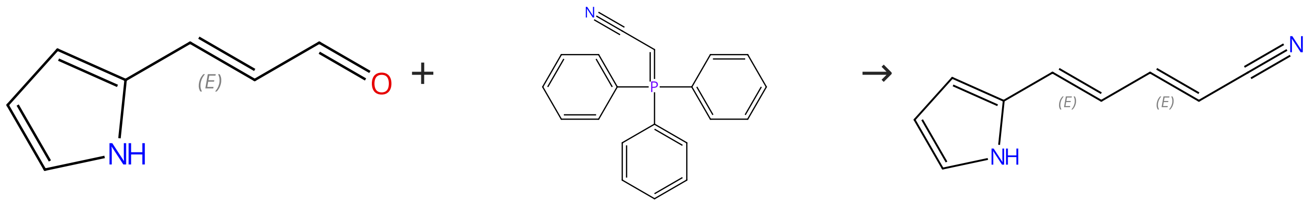
1.2 **Reagents:** Triethylamine; 1 h, 0 °C

Synthesis of Allenyl Esters by Horner- Wadsworth-Emmons Reactions of Ketenes Mediated by Isopropylmagnesium Bromide

By: Sano, Shigeki; et al

Synlett (2015), 26(15), 2135-2138.

Scheme 2 (1 Reaction) Steps: **1** Yield: **100%**



Double bond geometry shown

Suppliers (16) Suppliers (68)

Double bond geometry shown

31-339-CAS-23534722 Steps: **1** Yield: **100%**

1.1 **Solvents:** Toluene; 20 h, reflux

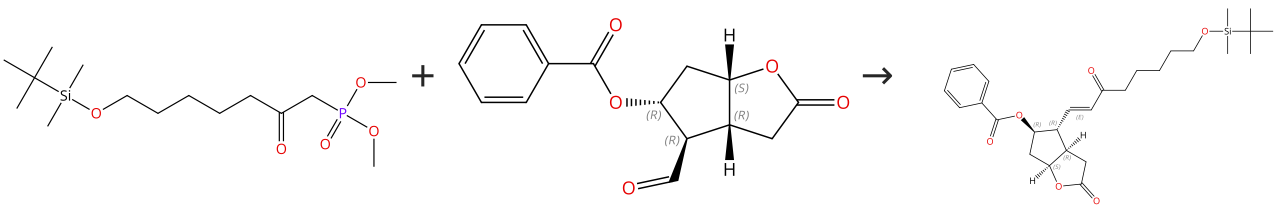
Experimental Protocols

Synthesis of polyenylpyrrole derivatives with selective growth inhibitory activity against T-cell acute lymphoblastic leukemia cells

By: Yoshida, Chihiro; et al

Bioorganic & Medicinal Chemistry Letters (2021), 37, 127837.

Scheme 3 (1 Reaction) Steps: **1** Yield: **100%**



Absolute stereochemistry shown, Rotation (-)

Absolute stereochemistry shown
Double bond geometry shown

Supplier (1) Suppliers (34)

31-339-CAS-9305489 Steps: **1** Yield: **100%**

1.1 **Reagents:** Sodium hydride
Solvents: Tetrahydrofuran; 0 °C; 1 h, rt; rt → 0 °C

1.2 2.5 h, rt

1.3 **Reagents:** Acetic acid; rt

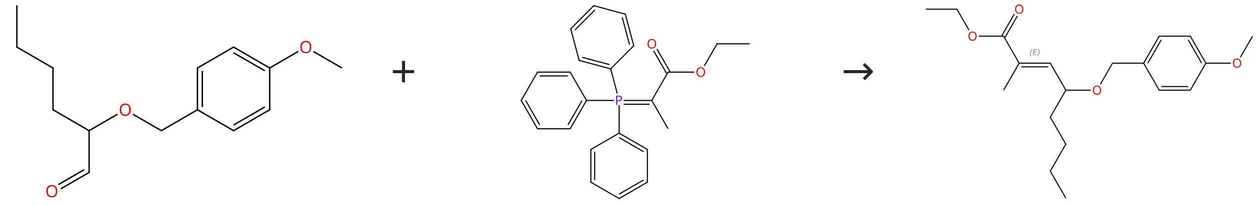
Experimental Protocols

Design and synthesis of novel prostaglandin E_2 ethanolamide and glycerol ester probes for the putative prostamide receptor(s)

By: Shelnut, Erin L.; et al

Tetrahedron Letters (2015), 56(11), 1411-1415.

Scheme 4 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (2)

Suppliers (80)

Double bond geometry shown

31-339-CAS-15622754 Steps: **1** Yield: **100%**

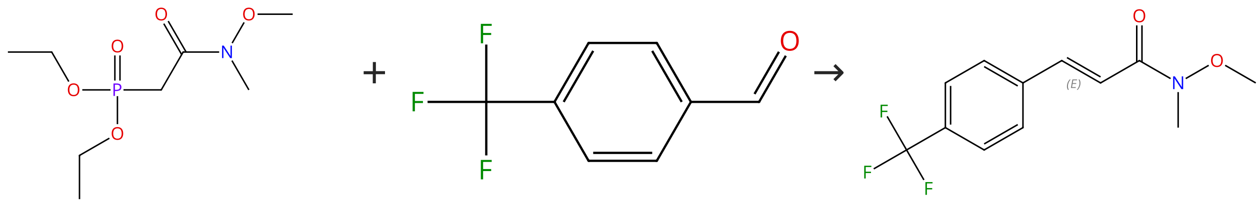
1.1 **Solvents:** Toluene; 100 °C

Synthesis and absolute configuration of formosusin A, a specific inhibitor of mammalian DNA polymerase β

By: Yajima, Arata; et al

Tetrahedron Letters (2016), 57(18), 2012-2015.

Scheme 5 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (65)

Suppliers (84)

Double bond geometry shown

Suppliers (18)

31-614-CAS-41961922 Steps: **1** Yield: **100%**

1.1 **Reagents:** Chloro(1-methylethyl)magnesium
Solvents: Tetrahydrofuran; 30 min, -78 °C

1.2 **Solvents:** Tetrahydrofuran; 12 h, rt

1.3 **Reagents:** Ammonium chloride
Solvents: Water; 0 °C

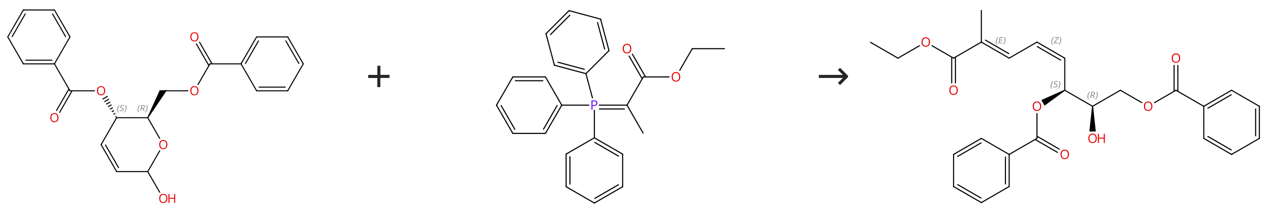
Experimental Protocols

The (E)-Selective Weinreb Amide-Type Horner-Wadsworth-Emmons Reaction: Effect of Reaction Conditions, Substrate Scope, Isolation of a Reactive Magnesium Phosphonoenolate, and Applications

By: Murata, Takatsugu; et al

Journal of Organic Chemistry (2024), 89(21), 15414-15435.

Scheme 6 (1 Reaction) Steps: **1** Yield: **100%**



Absolute stereochemistry shown

Suppliers (80)

Absolute stereochemistry shown, Rotation (-)
Double bond geometry shown

31-614-CAS-31753129 Steps: **1** Yield: **100%**

1.1 **Solvents:** Toluene; 2 h, 80 °C

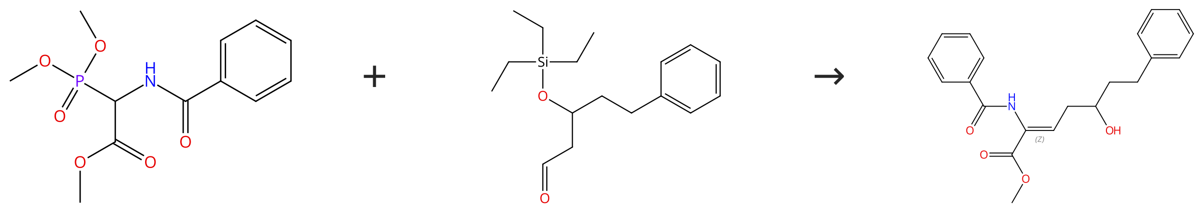
Experimental Protocols

Biomimetic Construction of a syn-2,7-Dimethyloxepane Ring via 7-Endo Cyclization

By: Sakai, Takeo; et al

Journal of Organic Chemistry (2022), 87(1), 579-594.

Scheme 7 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (6)

Double bond geometry shown

31-049-CAS-18505251 Steps: **1** Yield: **100%**

1.1 **Reagents:** 1,8-Diazabicyclo[5.4.0]undec-7-ene
Solvents: Dichloromethane; 20 min, 0 °C

1.2 **Solvents:** Dichloromethane; 0 °C

1.3 **Reagents:** Sulfuric acid
Solvents: Dichloromethane, Water; 4 h, 0 °C

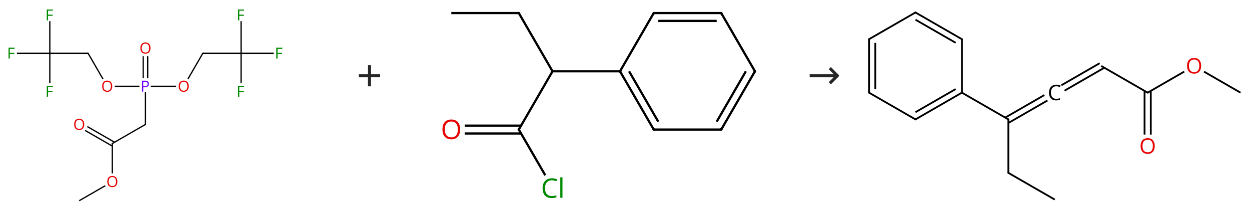
Experimental Protocols

An intramolecular oxa-Michael reaction on α,β-unsaturated α-amino-δ-hydroxycarboxylic acid esters. Synthesis of functio nalized 1,3-dioxanes

By: Becerra-Figueroa, L.; et al

Organic & Biomolecular Chemistry (2018), 16(8), 1277-1286.

Scheme 8 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (79) Suppliers (66) Suppliers (3)

31-339-CAS-8080072 Steps: **1** Yield: **100%**

1.1 **Reagents:** Bromo(1-methylethyl)magnesium
Solvents: Tetrahydrofuran; 1 h, 0 °C

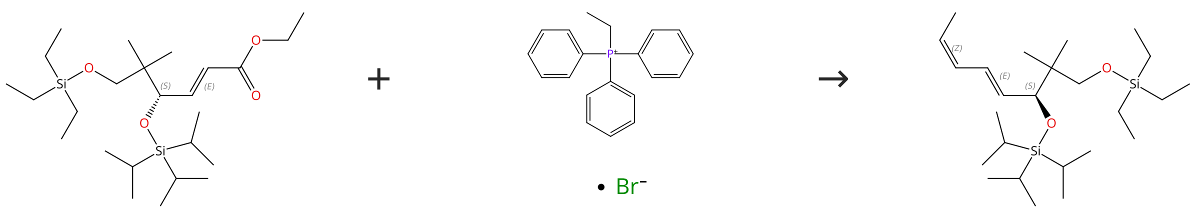
1.2 **Reagents:** Triethylamine; 1 h, 0 °C

Synthesis of Allenyl Esters by Horner- Wadsworth-Emmons Reactions of Ketenes Mediated by Isopropylmagnesium Bromide

By: Sano, Shigeki; et al

Synlett (2015), 26(15), 2135-2138.

Scheme 9 (1 Reaction) Steps: **1** Yield: **100%**



Absolute stereochemistry shown, Rotation (-)
Double bond geometry shown

Suppliers (80)

Absolute stereochemistry shown, Rotation (-)
Double bond geometry shown

31-339-CAS-23500286 | Steps: **1** Yield: **100%**

1.1 **Reagents:** Diisobutylaluminum hydride
Solvents: Dichloromethane, Hexane; 1 h, -78 °C; 10 min, -78 °C

1.2 **Reagents:** Potassium sodium tartrate
Solvents: Water; 12 h, -78 °C

1.3 **Reagents:** Manganese oxide (MnO_2)
Solvents: Dichloromethane; 22 h, rt

1.4 **Reagents:** Sodium bis(trimethylsilyl)amide
Solvents: Tetrahydrofuran; 15 min, 0 °C; 15 min, 0 °C

1.5 **Solvents:** Tetrahydrofuran; 30 min, 0 °C

1.6 **Reagents:** Ammonium chloride
Solvents: Water

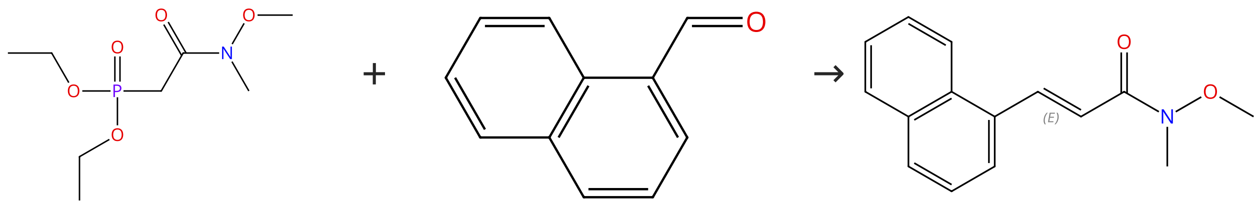
Experimental Protocols

Towards the total synthesis of aurodox: Prepar ation of the key hemiacetal-lactone

By: Ohara, Motoyoshi; et al

Tetrahedron Letters (2021), 66, 152799.

Scheme 10 (1 Reaction) | Steps: **1** Yield: **100%**



Suppliers (65)

Suppliers (83)

Double bond geometry shown

Supplier (1)

31-614-CAS-41961939 | Steps: **1** Yield: **100%**

1.1 **Reagents:** Chloro(1-methylethyl)magnesium
Solvents: Tetrahydrofuran; 30 min, -78 °C

1.2 **Solvents:** Tetrahydrofuran; 2 h, rt

1.3 **Reagents:** Ammonium chloride
Solvents: Water; 0 °C

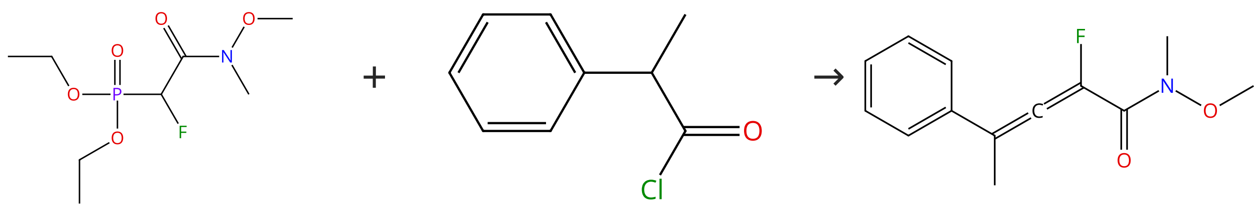
Experimental Protocols

The (E)-Selective Weinreb Amide-Type Horner-Wadsworth-Emmons Reaction: Effect of Reaction Condit ions, Substrate Scope, Isolation of a Reactive Magnesium Phosphonoenolate, and Applications

By: Murata, Takatsugu; et al

Journal of Organic Chemistry (2024), 89(21), 15414-15435.

Scheme 11 (1 Reaction) | Steps: **1** Yield: **100%**



Suppliers (37)

31-339-CAS-13752440 | Steps: **1** Yield: **100%**

1.1 **Reagents:** Bromo(1-methylethyl)magnesium
Solvents: Tetrahydrofuran; 1 h, 0 °C

1.2 **Reagents:** Triethylamine; 18 h, 0 °C

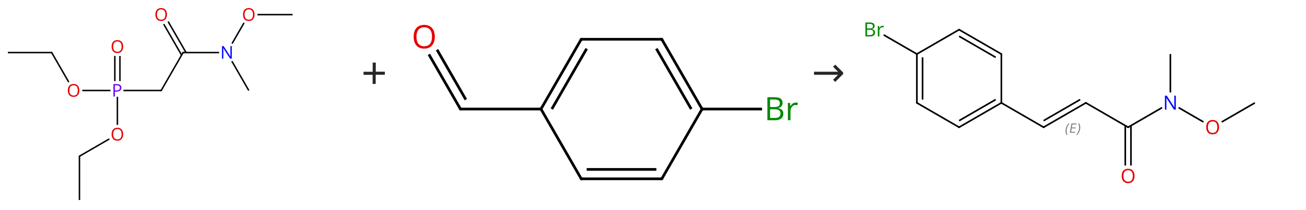
Synthesis of Allenyl Esters by Horner- Wadsworth-Emmons Reactions of Ketenes Mediated by Isopropylmagnesium Bromide

By: Sano, Shigeki; et al

Synlett (2015), 26(15), 2135-2138.

Scheme 12 (1 Reaction)

Steps: **1** Yield: **100%**



Suppliers (65)

Suppliers (91)

Double bond geometry shown

Suppliers (9)

31-614-CAS-41961941

Steps: **1** Yield: **100%**

1.1 **Reagents:** Chloro(1-methylethyl)magnesium
Solvents: Tetrahydrofuran; 30 min, -78 °C

1.2 **Solvents:** Tetrahydrofuran; 12 h, rt

1.3 **Reagents:** Ammonium chloride
Solvents: Water; 0 °C

Experimental Protocols

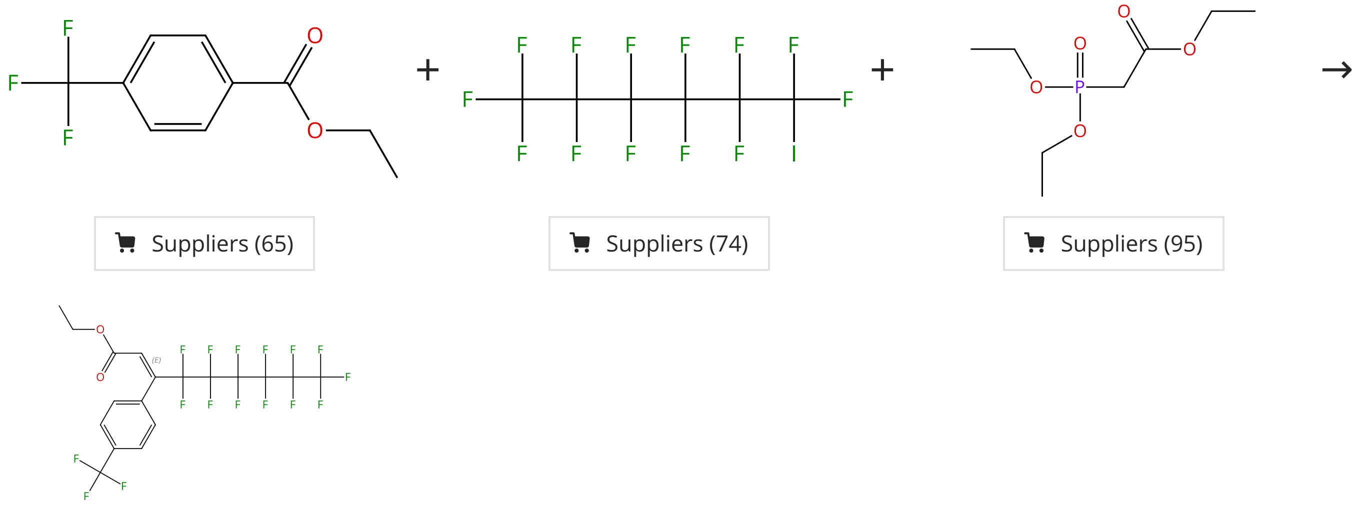
The (E)-Selective Weinreb Amide-Type Horner-Wadsworth-Emmons Reaction: Effect of Reaction Conditions, Substrate Scope, Isolation of a Reactive Magnesium Phosphonoenolate, and Applications

By: Murata, Takatsugu; et al

Journal of Organic Chemistry (2024), 89(21), 15414-15435.

Scheme 13 (1 Reaction)

Steps: **1** Yield: **99%**



Suppliers (65)

Suppliers (74)

Suppliers (95)

Double bond geometry shown

31-339-CAS-11799593

Steps: **1** Yield: **99%**

1.1 **Reagents:** Lithium, methyl-, compd. with lithium bromide (Li Br) (1:1)
Solvents: Diethyl ether; 30 min, - 80 °C

1.2 **Catalysts:** Diethylaluminum chloride
Solvents: Hexane; 15 min, -80 °C

1.3 **Reagents:** Triethylamine, Lithium bromide
Solvents: Tetrahydrofuran; 0 °C; 10 min, rt

1.4 rt; 1 h, rt

1.5 **Reagents:** Sodium fluoride
Solvents: Diethyl ether, Water; 0.5 h, rt

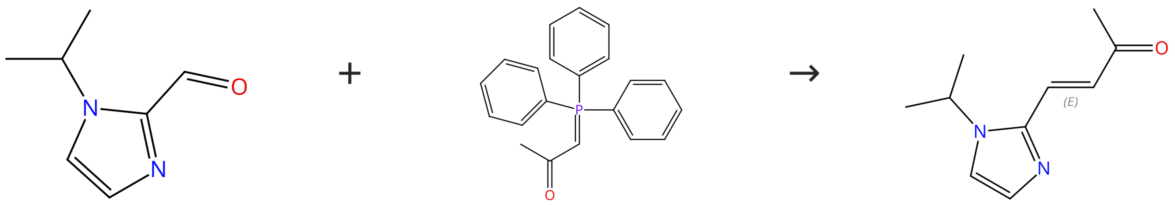
Experimental Protocols

Convenient stereoselective synthesis of β-perfluoroalkyl α,β-unsaturated esters via Horner-Wadsworth-Emmons reactions

By: Yamazaki, Takashi; et al

Tetrahedron (2015), 71(42), 8059-8066.

Scheme 14 (1 Reaction) Steps: **1** Yield: **99%**



Suppliers (55) Suppliers (85)

Double bond geometry shown

Suppliers (4)

31-339-CAS-3865720 Steps: **1** Yield: **99%**

1.1 **Solvents:** Toluene; 9 h, reflux

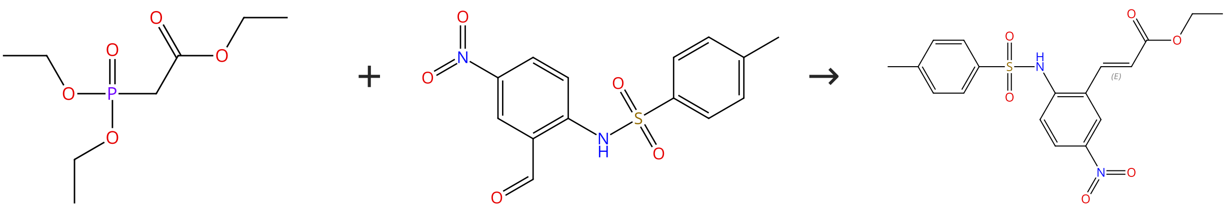
Experimental Protocols

Synthesis and evaluation of 1,7-diheteroarylhepta-1,4,6-trien-3-ones as curcumin-based anticancer agents

By: Wang, Rubing; et al

European Journal of Medicinal Chemistry (2016), 110, 164-180.

Scheme 15 (1 Reaction) Steps: **1** Yield: **99%**



Suppliers (95) Suppliers (9)

Double bond geometry shown

31-339-CAS-15968071 Steps: **1** Yield: **99%**

1.1 **Reagents:** 1,8-Diazabicyclo[5.4.0]undec-7-ene, Lithium bromide
Solvents: Acetonitrile; 0.5 h, rt

1.2 3 h, rt

1.3 **Reagents:** Potassium sodium tartrate
Solvents: Water

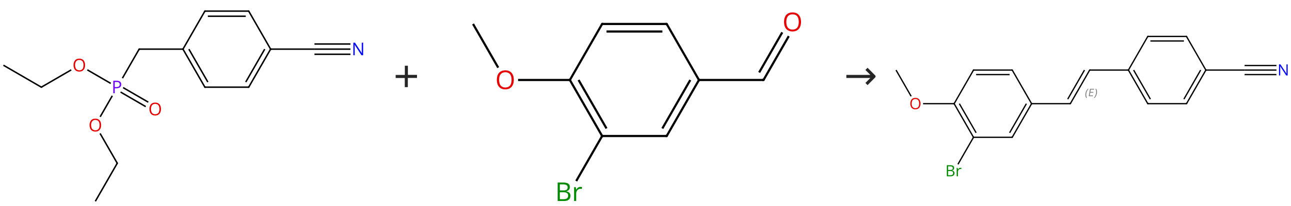
Experimental Protocols

Synthesis of 5-Amino-2,5-dihydro-1H-benzo[b]azepines Using a One-Pot Multibond Forming Process

By: Sharif, Salaheddin A. I.; et al

Journal of Organic Chemistry (2016), 81(15), 6697-6706.

Scheme 16 (1 Reaction) Steps: **1** Yield: **99%**



Suppliers (56) Suppliers (82)

Double bond geometry shown

31-614-CAS-40326729 Steps: **1** Yield: **99%**

1.1 **Reagents:** Sodium hydride
Solvents: Tetrahydrofuran; 0 °C; 30 min, 0 °C

1.2 **Solvents:** Tetrahydrofuran; 0 °C; 0 °C → rt; overnight, rt

1.3 **Solvents:** Water; rt

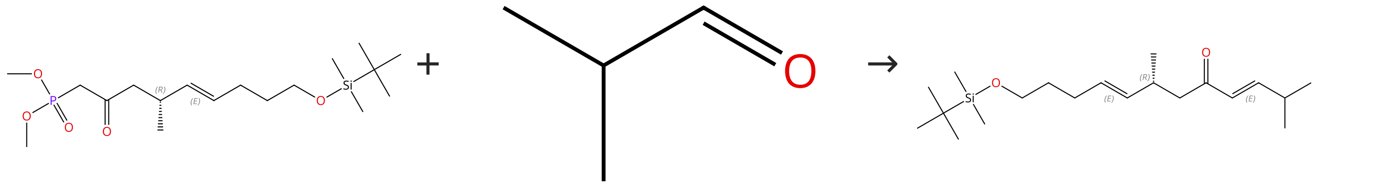
Experimental Protocols

Catalytic 1,1-diazidation of alkenes

By: Qiu, Wangzhen; et al

Nature Communications (2024), 15(1), 3632.

Scheme 17 (2 Reactions) Steps: **1** Yield: **99%**



Absolute stereochemistry shown, Rotation (-) Double bond geometry shown

Suppliers (74)

Absolute stereochemistry shown, Rotation (-) Double bond geometry shown

31-339-CAS-16194445 Steps: **1** Yield: **99%**

1.1 **Reagents:** Diisopropylethylamine, Lithium chloride
Solvents: Acetonitrile; 15 h, rt

1.2 **Reagents:** Ammonium chloride
Solvents: Water

Experimental Protocols

Stereodivergent Synthesis and Configur ational Assignment of the C1-C15 Segment of Amphirionin-5

By: Kanto, Moemi; et al

Journal of Organic Chemistry (2016), 81(19), 9105-9121.

31-339-CAS-5699259 Steps: **1** Yield: **99%**

1.1 **Reagents:** Diisopropylethylamine, Lithium chloride
Solvents: Acetonitrile; 15 h, rt

1.2 **Reagents:** Ammonium chloride
Solvents: Water; rt

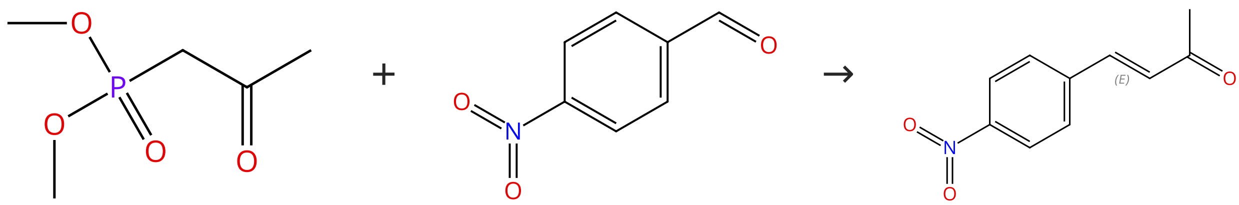
Experimental Protocols

Synthetic Studies on Amphirionin-5: Stereochemical Assignme nt/Reassignment of the C1-C9 Portion through Stereodi vergent Synthesis

By: Kanto, Moemi; et al

Organic Letters (2016), 18(1), 112-115.

Scheme 18 (1 Reaction) Steps: **1** Yield: **99%**



Suppliers (75)

Suppliers (92)

Double bond geometry shown

Suppliers (29)

31-339-CAS-4402293 Steps: **1** Yield: **99%**

1.1 **Reagents:** Sodium hydroxide
Solvents: Water; rt

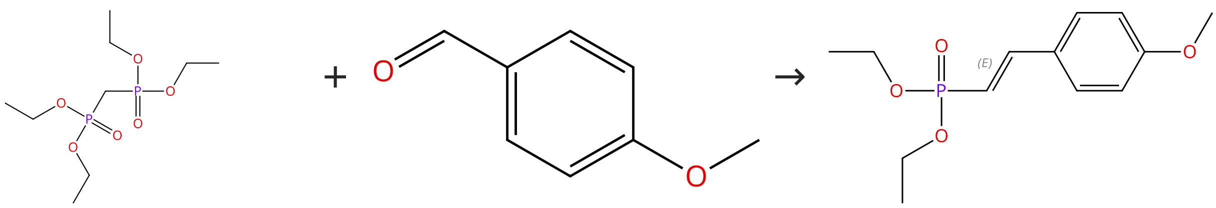
Experimental Protocols

Efficient synthesis of α,β-unsaturated ketones with trans-selective Horner-Wadsworth-Emmons reaction in water

By: Jafari, Abbas Ali; et al

Environmental Chemistry Letters (2016), 14(2), 223-228.

Scheme 19 (1 Reaction) Steps: **1** Yield: **99%**



Suppliers (70)

Suppliers (128)

Double bond geometry shown

Suppliers (4)

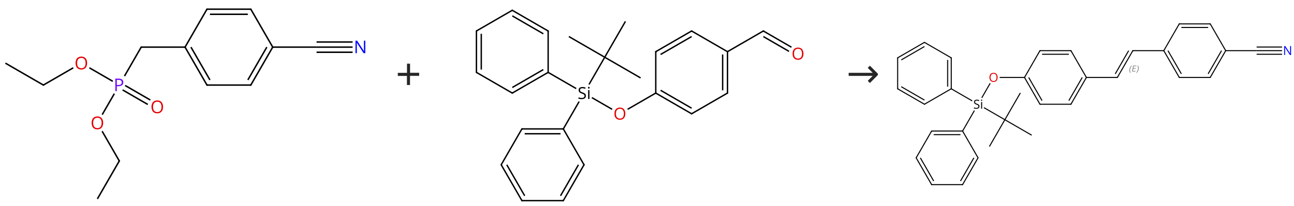
31-614-CAS-34373366	Steps: **1** Yield: **99%**
1.1 **Reagents:** Potassium *tert*-butoxide; 5 min, rt	
Experimental Protocols	

Mechanosynthesis of phosphonocinnamic esters through solvent-free Horner-Wadsworth-Emmons reaction

By: Jimenez-Arellanes, Maria Adelina; et al

Phosphorus, Sulfur and Silicon and the Related Elements (2022), 197(10), 1045-1054.

Scheme 20 (1 Reaction) Steps: **1** Yield: **99%**



Suppliers (56)　Suppliers (8)

Double bond geometry shown

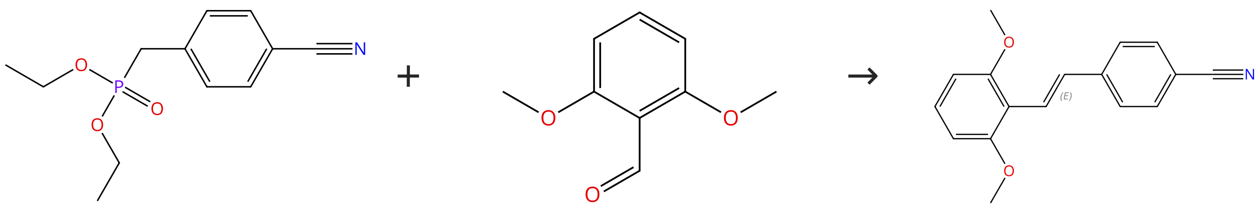
31-614-CAS-40326725	Steps: **1** Yield: **99%**
1.1 **Reagents:** Sodium hydride **Solvents:** Tetrahydrofuran; 0 °C; 30 min, 0 °C	
1.2 **Solvents:** Tetrahydrofuran; 0 °C; 0 °C → rt; overnight, rt	
1.3 **Solvents:** Water; rt	
Experimental Protocols	

Catalytic 1,1-diazidation of alkenes

By: Qiu, Wangzhen; et al

Nature Communications (2024), 15(1), 3632.

Scheme 21 (1 Reaction) Steps: **1** Yield: **99%**



Suppliers (56)　Suppliers (99)

Double bond geometry shown

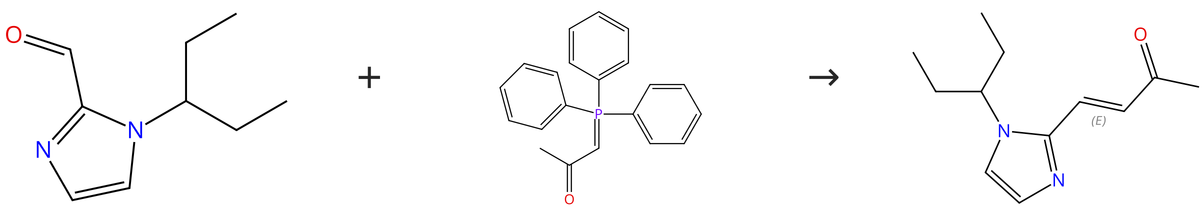
31-614-CAS-40326738	Steps: **1** Yield: **99%**
1.1 **Reagents:** Sodium hydride **Solvents:** Tetrahydrofuran; 0 °C; 30 min, 0 °C	
1.2 **Solvents:** Tetrahydrofuran; 0 °C; 0 °C → rt; overnight, rt	
1.3 **Solvents:** Water; rt	
Experimental Protocols	

Catalytic 1,1-diazidation of alkenes

By: Qiu, Wangzhen; et al

Nature Communications (2024), 15(1), 3632.

Scheme 22 (1 Reaction) Steps: **1** Yield: **99%**



Suppliers (5)　Suppliers (85)

Double bond geometry shown

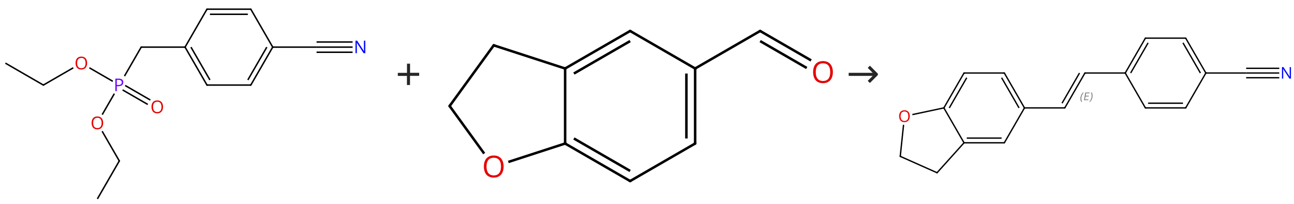
31-339-CAS-14806338	Steps: **1** Yield: **99%**
1.1 **Solvents:** Toluene; 9 h, reflux	
Experimental Protocols	

Synthesis and evaluation of 1,7-diheteroarylhepta-1,4,6-trien-3-ones as curcumin-based anticancer agents

By: Wang, Rubing; et al

European Journal of Medicinal Chemistry (2016), 110, 164-180.

Scheme 23 (1 Reaction) Steps: **1** Yield: **99%**



Suppliers (56) Suppliers (86)

Double bond geometry shown

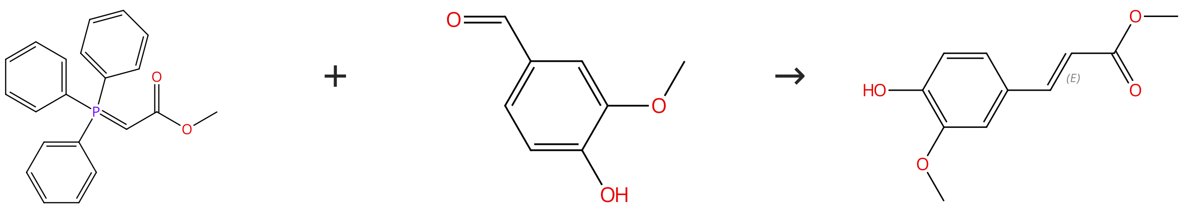
31-614-CAS-40326736	Steps: **1** Yield: **99%**
1.1 **Reagents:** Sodium hydride **Solvents:** Tetrahydrofuran; 0 °C; 30 min, 0 °C	
1.2 **Solvents:** Tetrahydrofuran; 0 °C; 0 °C → rt; overnight, rt	
1.3 **Solvents:** Water; rt	
Experimental Protocols	

Catalytic 1,1-diazidation of alkenes

By: Qiu, Wangzhen; et al

Nature Communications (2024), 15(1), 3632.

Scheme 24 (1 Reaction) Steps: **1** Yield: **99%**



Suppliers (86) Suppliers (146)

Double bond geometry shown

Suppliers (71)

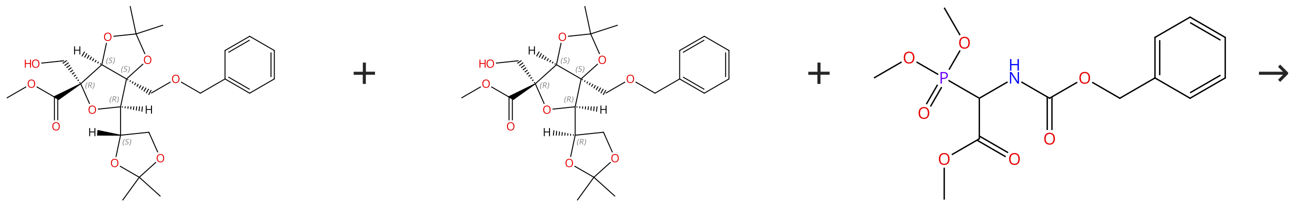
31-339-CAS-19339813	Steps: **1** Yield: **99%**
1.1 **Solvents:** Water; 0.5 h, 90 °C	
Experimental Protocols	

Synthesis and antioxidant activity of caffeic acid derivatives

By: Sidoryk, Katarzyna; et al

Molecules (2018), 23(9), 2199/1-2199/12.

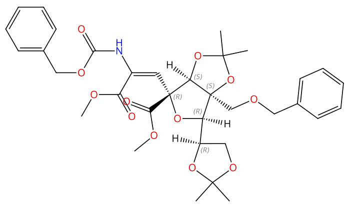
Scheme 25 (1 Reaction) Steps: **1** Yield: **98%**



Absolute stereochemistry shown

Absolute stereochemistry shown

Suppliers (99)



Absolute stereochemistry shown
Double bond geometry unknown

31-339-CAS-21081568 Steps: **1** Yield: **98%**

1.1 **Reagents:** Diisopropylethylamine, Sulfur trioxide-pyridine
Solvents: Dimethyl sulfoxide

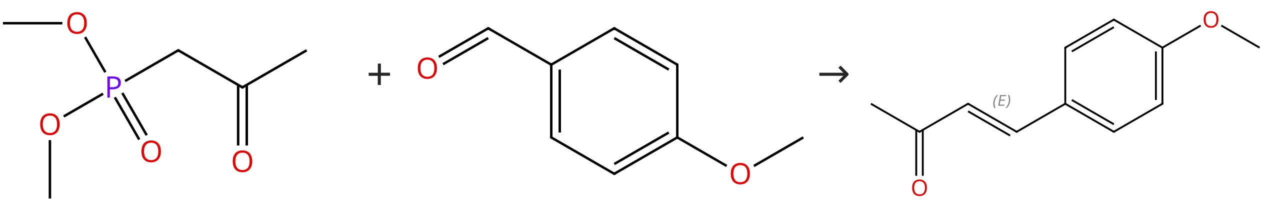
1.2 **Reagents:** 1,1,3,3-Tetramethylguanidine
Solvents: Dichloromethane

A monocyclic neodysiherbaine analog: Synthesis and evaluation

By: Fukushima, Koichi; et al

Bioorganic & Medicinal Chemistry Letters (2016), 26(21), 5164-5167.

Scheme 26 (1 Reaction) Steps: **1** Yield: **98%**



Suppliers (75)

Suppliers (128)

Double bond geometry shown

Suppliers (43)

31-339-CAS-10748065 Steps: **1** Yield: **98%**

1.1 **Reagents:** Sodium hydroxide
Solvents: Water; 3 h; rt

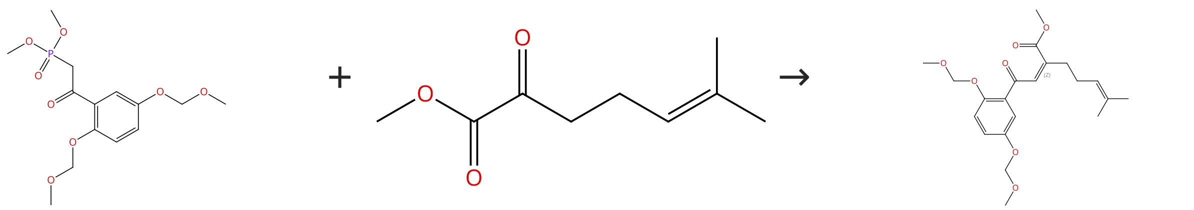
Experimental Protocols

Efficient synthesis of α,β-unsaturated ketones with trans-selective Horner-Wadsworth-Emmons reaction in water

By: Jafari, Abbas Ali; et al

Environmental Chemistry Letters (2016), 14(2), 223-228.

Scheme 27 (1 Reaction) Steps: **1** Yield: **98%**



Suppliers (4)

Double bond geometry shown

31-614-CAS-42509292 Steps: **1** Yield: **98%**

1.1 **Reagents:** Sodium *tert*-butoxide
Solvents: Tetrahydrofuran; 0 °C

1.2 **Reagents:** Ammonium chloride
Solvents: Water; 0 °C

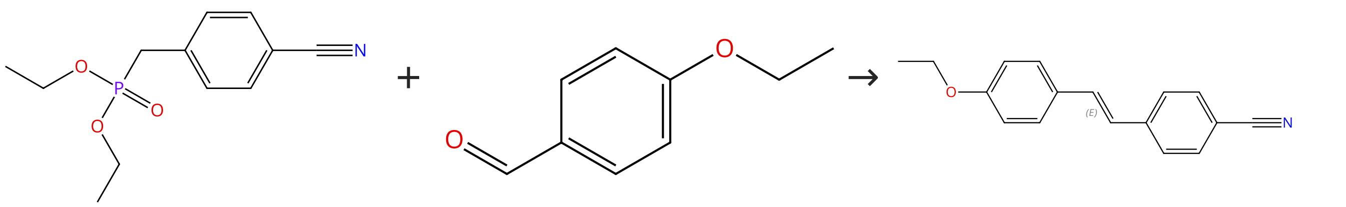
Experimental Protocols

Synthesis and Biological Activity of Tetrac yclic Dispiro Core Derivatives of Natural Products Dispirocochlearoids A-C

By: Pei, Junping; et al

Organic Letters (2024), 26(45), 9654-9658.

Scheme 28 (1 Reaction) Steps: **1** Yield: **98%**



Suppliers (56)

Suppliers (89)

Double bond geometry shown

Suppliers (7)

31-614-CAS-40326723 Steps: **1** Yield: **98%**

1.1 **Reagents:** Sodium hydride
Solvents: Tetrahydrofuran; 0 °C; 30 min, 0 °C

1.2 **Solvents:** Tetrahydrofuran; 0 °C; 0 °C → rt; overnight, rt

1.3 **Solvents:** Water; rt

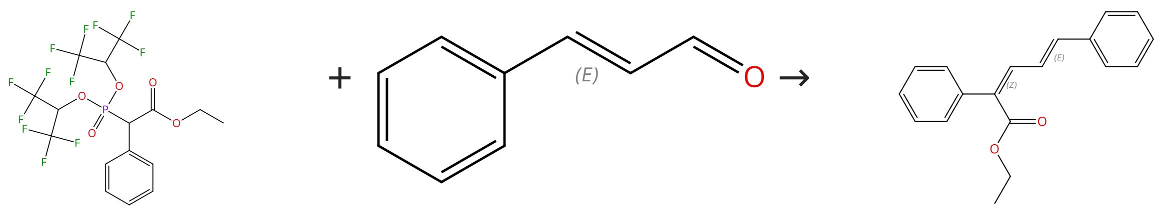
Experimental Protocols

Catalytic 1,1-diazidation of alkenes

By: Qiu, Wangzhen; et al

Nature Communications (2024), 15(1), 3632.

Scheme 29 (1 Reaction) Steps: **1** Yield: **98%**



Double bond geometry shown

Suppliers (109)

Double bond geometry shown

Supplier (1)

31-614-CAS-45379288 Steps: **1** Yield: **98%**

1.1 **Reagents:** Sodium hydride
Solvents: Tetrahydrofuran; -78 °C; -78 °C → rt; 48 h, rt

1.2 **Reagents:** Ammonium chloride
Solvents: Water; rt

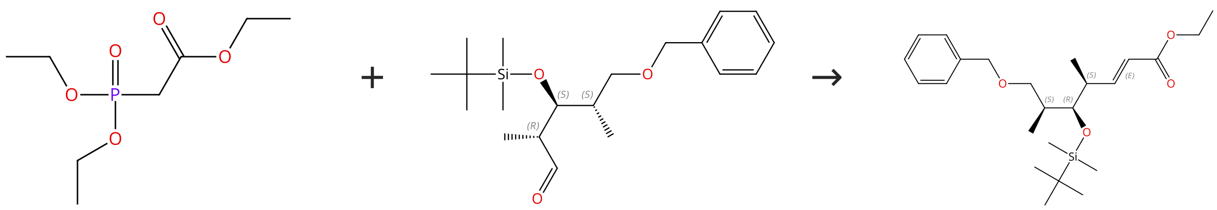
Experimental Protocols

Z-Selective Synthesis of Trisubstituted Alkenes by the HWE Reaction Using Modified Still-Gennari Type Reagents

By: Janicki, Ignacy

Journal of Organic Chemistry (2025), 90(16), 5725-5728.

Scheme 30 (1 Reaction) Steps: **1** Yield: **98%**



Suppliers (95)

Absolute stereochemistry shown, Rotation (-)

Absolute stereochemistry shown, Rotation (-)
Double bond geometry shown

31-339-CAS-4103170	Steps: **1** Yield: **98%**
1.1	**Reagents:** Sodium hydride **Solvents:** Tetrahydrofuran; 30 min, 0 °C
1.2	**Solvents:** Tetrahydrofuran; 0 °C → - 78 °C; 1 h, 0 °C
1.3	**Solvents:** Diethyl ether, Water

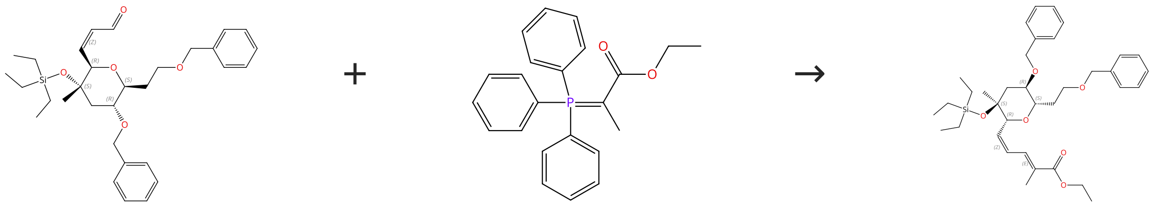
Experimental Protocols

Asymmetric synthesis of emericellamide B

By: Ren, Rong-Guo; et al

Chinese Chemical Letters (2015), 26(10), 1209-1215.

Scheme 31 (1 Reaction) Steps: **1** Yield: **98%**



Absolute stereochemistry shown, Rotation (-) Double bond geometry shown

Suppliers (80)

Absolute stereochemistry shown, Rotation (-) Double bond geometry shown

31-614-CAS-31753189	Steps: **1** Yield: **98%**
1.1	**Solvents:** Dichloromethane; 18 h, rt

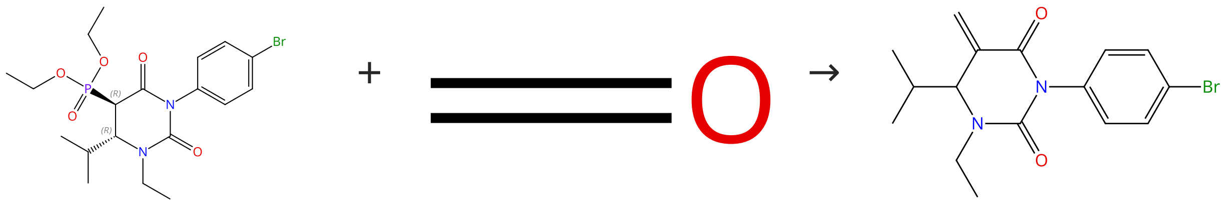
Experimental Protocols

Biomimetic Construction of a syn-2,7-Dimethyloxepane Ring via 7-Endo Cyclization

By: Sakai, Takeo; et al

Journal of Organic Chemistry (2022), 87(1), 579-594.

Scheme 32 (1 Reaction) Steps: **1** Yield: **98%**



Relative stereochemistry shown

Suppliers (188)

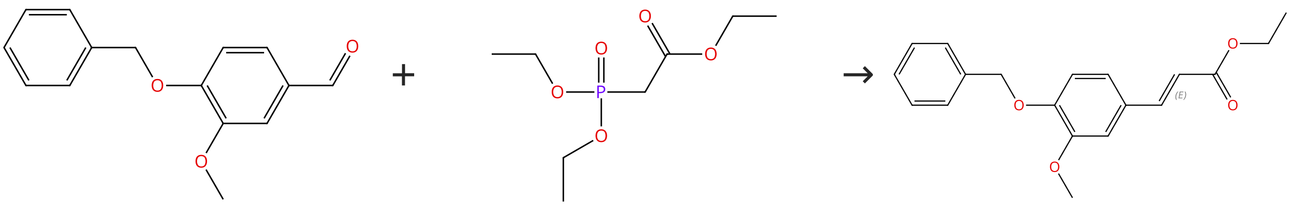
31-339-CAS-20242527	Steps: **1** Yield: **98%**
1.1	**Reagents:** Sodium hydride **Solvents:** Tetrahydrofuran; 5 min, rt
1.2	60 min, rt

Enantioselective synthesis of 5-methylidenedihydrouracils as potential anticancer agents

By: Pieta, Marlena; et al

Tetrahedron (2019), 75(17), 2495-2505.

Scheme 33 (1 Reaction) Steps: **1** Yield: **98%**



Suppliers (80)

Suppliers (95)

Double bond geometry shown

Suppliers (52)

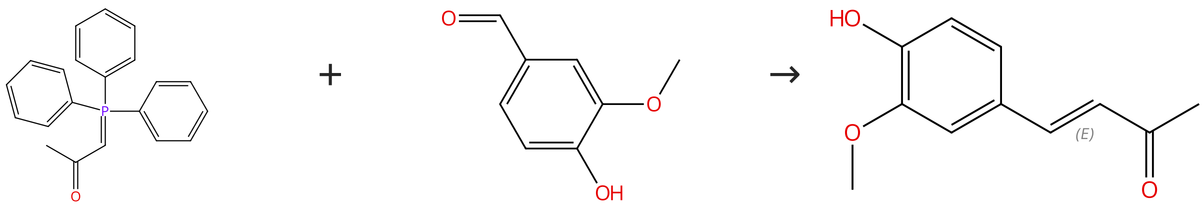
31-339-CAS-19339226	Steps: **1** Yield: **98%**
1.1	**Reagents:** Potassium carbonate **Catalysts:** 1,8-Diazabicyclo[5.4.0]undec-7-ene **Solvents:** Dimethylformamide, Dichloromethane; 24 - 36 h, rt
1.2	**Solvents:** Water; 30 min, cooled
Experimental Protocols	

Synthesis of cinnamic amide derivatives and their anti-melanogenic effect in α-MSH-stimulated B16F10 melanoma cells

By: Ullah, Sultan; et al

European Journal of Medicinal Chemistry (2019), 161, 78-92.

Scheme 34 (1 Reaction) Steps: **1** Yield: **98%**



Suppliers (85)

Suppliers (146)

Double bond geometry shown

Suppliers (46)

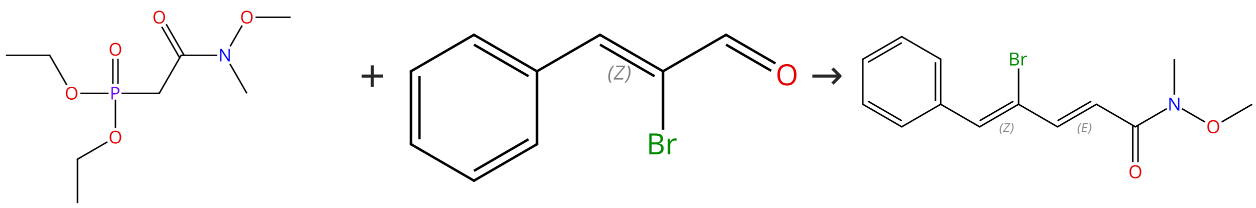
31-339-CAS-19337266	Steps: **1** Yield: **98%**
1.1	**Solvents:** Water; 0.5 h, 90 °C
Experimental Protocols	

Synthesis and antioxidant activity of caffeic acid derivatives

By: Sidoryk, Katarzyna; et al

Molecules (2018), 23(9), 2199/1-2199/12.

Scheme 35 (1 Reaction) Steps: **1** Yield: **98%**



Suppliers (65)

Double bond geometry shown

Suppliers (27)

Double bond geometry shown

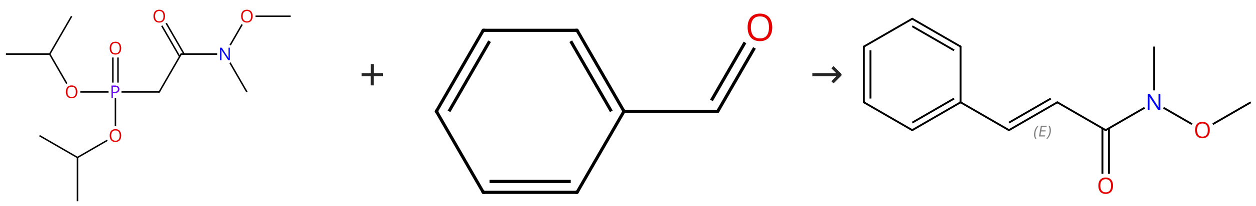
31-614-CAS-41961931	Steps: **1** Yield: **98%**
1.1	**Reagents:** Chloro(1-methylethyl)magnesium **Solvents:** Tetrahydrofuran; 30 min, -78 °C
1.2	**Solvents:** Tetrahydrofuran; 12 h, rt
1.3	**Reagents:** Ammonium chloride **Solvents:** Water; 0 °C
Experimental Protocols	

The (E)-Selective Weinreb Amide-Type Horner-Wadsworth-Emmons Reaction: Effect of Reaction Conditions, Substrate Scope, Isolation of a Reactive Magnesium Phosphonoenolate, and Applications

By: Murata, Takatsugu; et al

Journal of Organic Chemistry (2024), 89(21), 15414-15435.

Scheme 36 (1 Reaction) Steps: **1** Yield: **98%**



Suppliers (69)

Double bond geometry shown

Suppliers (26)

31-614-CAS-41961977	Steps: **1** Yield: **98%**
1.1	**Reagents:** Chloro(1-methylethyl)magnesium **Solvents:** Tetrahydrofuran; 30 min, -78 °C
1.2	**Solvents:** Tetrahydrofuran; 12 h, rt
1.3	**Reagents:** Ammonium chloride **Solvents:** Water; 0 °C

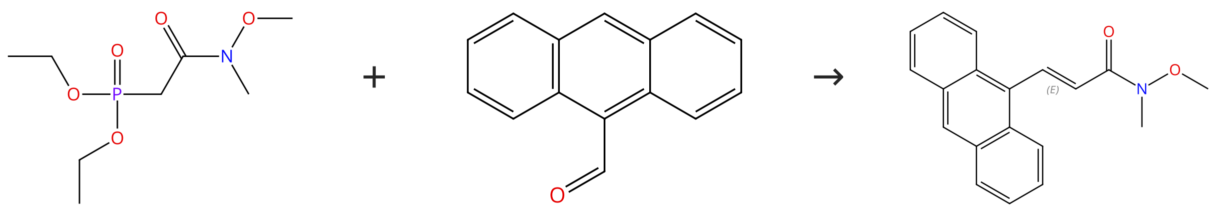
Experimental Protocols

The (E)-Selective Weinreb Amide-Type Horner-Wadsworth-Emmons Reaction: Effect of Reaction Conditions, Substrate Scope, Isolation of a Reactive Magnesium Phosphonoenolate, and Applications

By: Murata, Takatsugu; et al

Journal of Organic Chemistry (2024), 89(21), 15414-15435.

Scheme 37 (1 Reaction) Steps: **1** Yield: **98%**



Suppliers (65) Suppliers (92)

Double bond geometry shown

31-614-CAS-41961943	Steps: **1** Yield: **98%**
1.1	**Reagents:** Chloro(1-methylethyl)magnesium **Solvents:** Tetrahydrofuran; 30 min, -78 °C
1.2	**Solvents:** Tetrahydrofuran; 6 h, reflux
1.3	**Reagents:** Ammonium chloride **Solvents:** Water; 0 °C

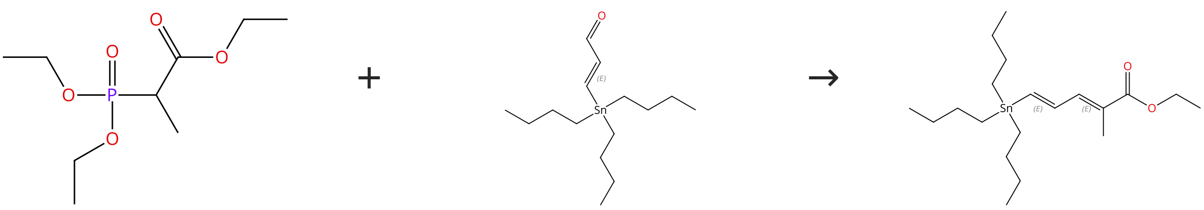
Experimental Protocols

The (E)-Selective Weinreb Amide-Type Horner-Wadsworth-Emmons Reaction: Effect of Reaction Conditions, Substrate Scope, Isolation of a Reactive Magnesium Phosphonoenolate, and Applications

By: Murata, Takatsugu; et al

Journal of Organic Chemistry (2024), 89(21), 15414-15435.

Scheme 38 (1 Reaction) Steps: **1** Yield: **98%**



Suppliers (79)

Double bond geometry shown

Double bond geometry shown

Suppliers (8)

31-339-CAS-20848623	Steps: **1** Yield: **98%**
1.1	**Reagents:** Butyllithium **Solvents:** Tetrahydrofuran; 0.5 h, 0 °C
1.2	**Solvents:** Tetrahydrofuran; 3 h, 0 °C

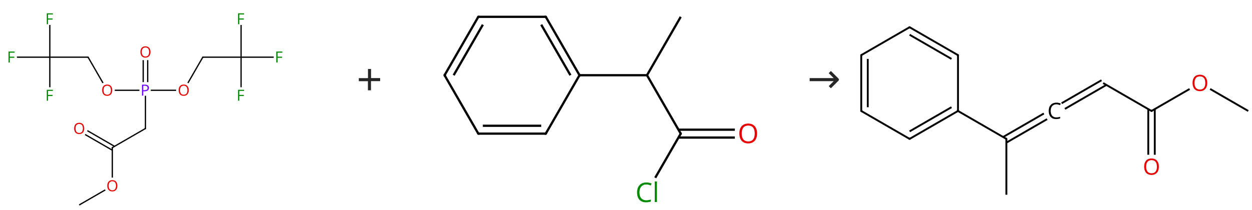
Experimental Protocols

Improved synthesis of key fragments for the preparation of natural product incednine

By: Sarceda, Sandra; et al

Tetrahedron (2019), 75(43), 130604.

Scheme 39 (1 Reaction) Steps: **1** Yield: **98%**



Suppliers (79) Suppliers (37) Suppliers (3)

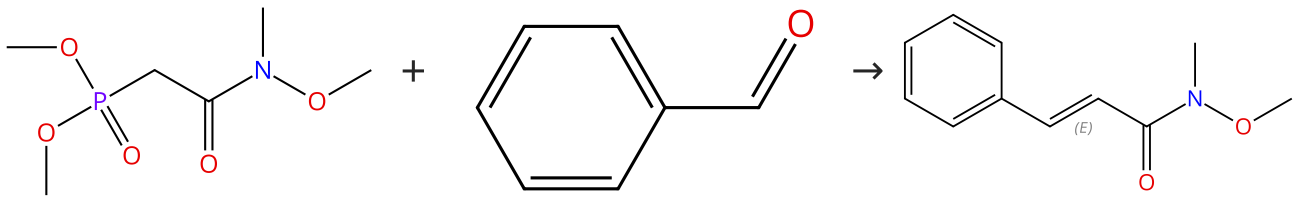
31-339-CAS-4140992	Steps: **1** Yield: **98%**
1.1	**Reagents:** Bromo(1-methylethyl)magnesium **Solvents:** Tetrahydrofuran; 1 h, 0 °C
1.2	**Reagents:** Triethylamine; 1 h, 0 °C

Synthesis of Allenyl Esters by Horner- Wadsworth-Emmons Reactions of Ketenes Mediated by Isopropylmagnesium Bromide

By: Sano, Shigeki; et al

Synlett (2015), 26(15), 2135-2138.

Scheme 40 (1 Reaction) Steps: **1** Yield: **98%**



Suppliers (69)

Double bond geometry shown

Suppliers (26)

31-614-CAS-41961974	Steps: **1** Yield: **98%**
1.1	**Reagents:** Chloro(1-methylethyl)magnesium **Solvents:** Tetrahydrofuran; 30 min, -78 °C
1.2	**Solvents:** Tetrahydrofuran; 12 h, rt
1.3	**Reagents:** Ammonium chloride **Solvents:** Water; 0 °C

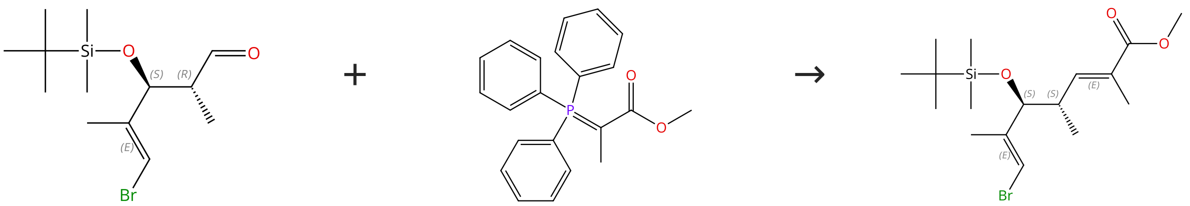
Experimental Protocols

The (E)-Selective Weinreb Amide-Type Horner-Wadsworth-Emmons Reaction: Effect of Reaction Conditions, Substrate Scope, Isolation of a Reactive Magnesium Phosphonoenolate, and Applications

By: Murata, Takatsugu; et al

Journal of Organic Chemistry (2024), 89(21), 15414-15435.

Scheme 41 (1 Reaction) Steps: **1** Yield: **98%**



Absolute stereochemistry shown, Rotation (-) Double bond geometry shown

Suppliers (54)

Absolute stereochemistry shown, Rotation (-) Double bond geometry shown

31-614-CAS-37110009	Steps: **1** Yield: **98%**
1.1	**Solvents:** Dichloromethane; 38 h, rt

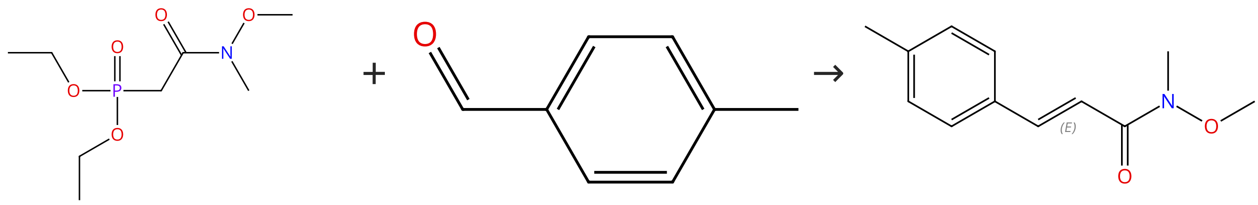
Experimental Protocols

Synthesis of the Macrolactone Cores of Maltepolides via a Diene-Ene Ring-Closing Metathesis Strategy

By: Sit, Man Ki; et al

Organic Letters (2023), 25(10), 1633-1637.

Scheme 42 (1 Reaction) Steps: **1** Yield: **98%**



Suppliers (65)

Suppliers (104)

Double bond geometry shown

Suppliers (6)

31-614-CAS-41961923	Steps: **1** Yield: **98%**
1.1	**Reagents:** Chloro(1-methylethyl)magnesium **Solvents:** Tetrahydrofuran; 30 min, -78 °C
1.2	**Solvents:** Tetrahydrofuran; 12 h, rt
1.3	**Reagents:** Ammonium chloride **Solvents:** Water; 0 °C

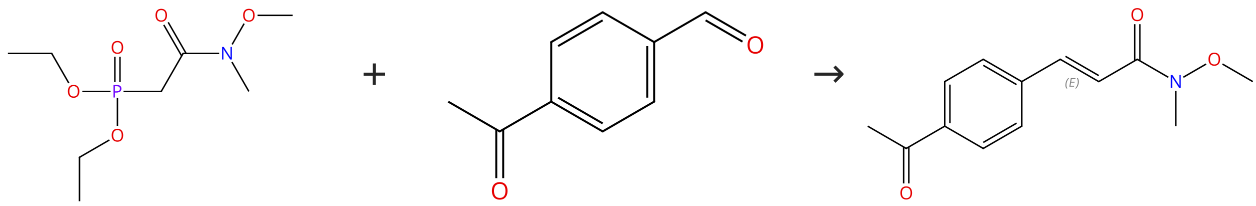
Experimental Protocols

The (E)-Selective Weinreb Amide-Type Horner-Wadsworth-Emmons Reaction: Effect of Reaction Conditions, Substrate Scope, Isolation of a Reactive Magnesium Phosphonoenolate, and Applications

By: Murata, Takatsugu; et al

Journal of Organic Chemistry (2024), 89(21), 15414-15435.

Scheme 43 (1 Reaction) Steps: **1** Yield: **98%**



Suppliers (65)

Suppliers (66)

Double bond geometry shown

31-614-CAS-41961955	Steps: **1** Yield: **98%**
1.1	**Reagents:** Chloro(1-methylethyl)magnesium **Solvents:** Tetrahydrofuran; 30 min, -78 °C
1.2	**Solvents:** Tetrahydrofuran; 12 h, rt
1.3	**Reagents:** Ammonium chloride **Solvents:** Water; 0 °C

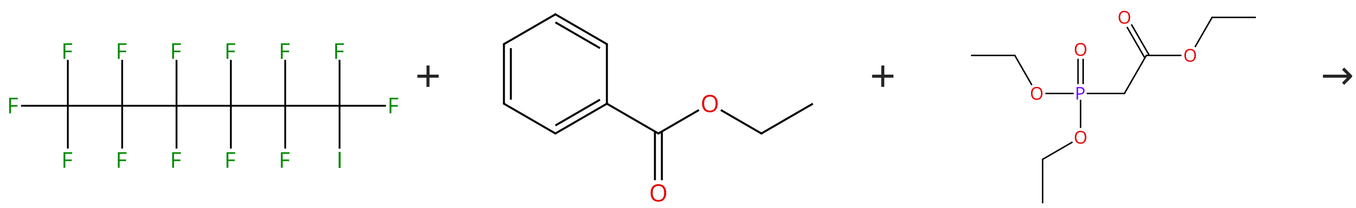
Experimental Protocols

The (E)-Selective Weinreb Amide-Type Horner-Wadsworth-Emmons Reaction: Effect of Reaction Conditions, Substrate Scope, Isolation of a Reactive Magnesium Phosphonoenolate, and Applications

By: Murata, Takatsugu; et al

Journal of Organic Chemistry (2024), 89(21), 15414-15435.

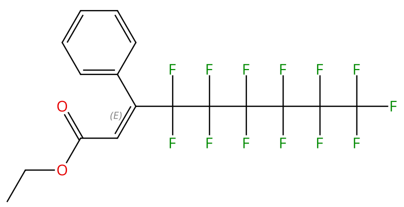
Scheme 44 (1 Reaction) Steps: **1** Yield: **98%**



Suppliers (74)

Suppliers (90)

Suppliers (95)



Double bond geometry shown

31-339-CAS-872711 Steps: **1** Yield: **98%**

1.1 **Reagents:** Lithium, methyl-, compd. with lithium bromide (Li Br) (1:1)
Solvents: Diethyl ether; 30 min, - 80 °C

1.2 **Catalysts:** Diethylaluminum chloride
Solvents: Hexane; 15 min, -80 °C

1.3 **Reagents:** Triethylamine, Lithium bromide
Solvents: Tetrahydrofuran; 0 °C; 10 min, rt

1.4 rt; 1 h, rt

1.5 **Reagents:** Sodium fluoride
Solvents: Diethyl ether, Water; 0.5 h, rt

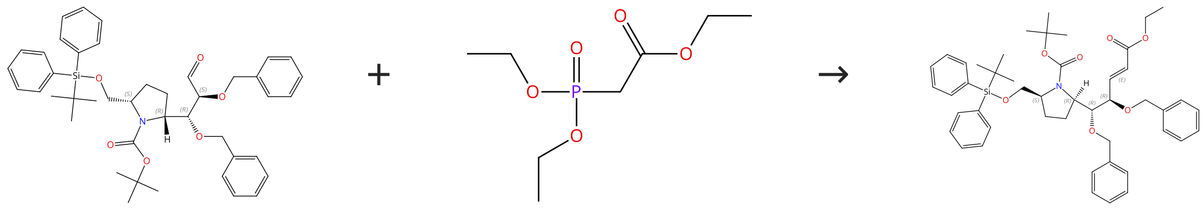
Experimental Protocols

Convenient stereoselective synthesis of β-perfluoroalkyl α,β-unsaturated esters via Horner-Wadsworth-Emmons reactions

By: Yamazaki, Takashi; et al

Tetrahedron (2015), 71(42), 8059-8066.

Scheme 45 (1 Reaction) Steps: **1** Yield: **98%**



Absolute stereochemistry shown

Suppliers (95)

Absolute stereochemistry shown, Rotation (+)
Double bond geometry shown

31-339-CAS-2084801 Steps: **1** Yield: **98%**

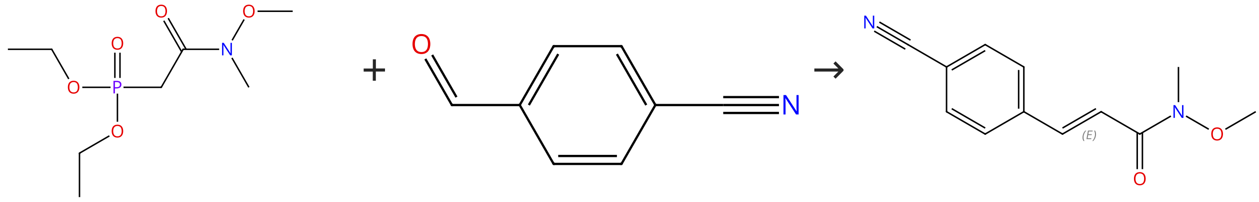
1.1 **Reagents:** Sodium hydride
Solvents: Tetrahydrofuran

Enantiodivergent strategy for the synthesis of polyhydr oxylated pyrrolizidines and evaluation of their inhibitory activities against glycosidases

By: Minehira, Daisuke; et al

Tetrahedron Letters (2015), 56(2), 331-334.

Scheme 46 (1 Reaction) Steps: **1** Yield: **98%**



Suppliers (65)

Suppliers (91)

Double bond geometry shown

Suppliers (7)

31-614-CAS-41961961 Steps: **1** Yield: **98%**

1.1 **Reagents:** Chloro(1-methylethyl)magnesium
Solvents: Tetrahydrofuran; 30 min, -78 °C

1.2 **Solvents:** Tetrahydrofuran; 12 h, rt

1.3 **Reagents:** Ammonium chloride
Solvents: Water; 0 °C

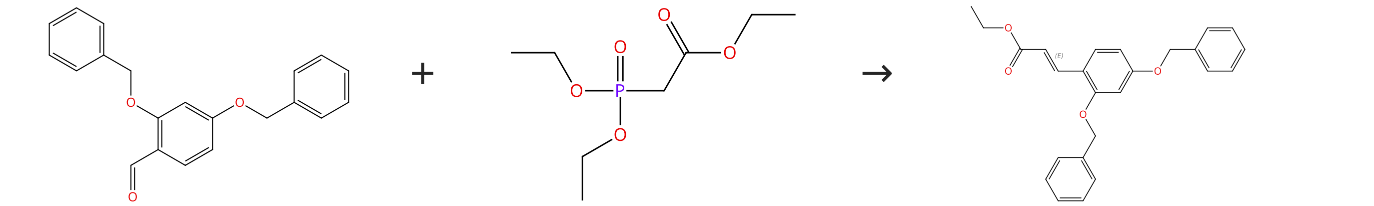
Experimental Protocols

The (E)-Selective Weinreb Amide-Type Horner-Wadsworth-Emmons Reaction: Effect of Reaction Conditions, Substrate Scope, Isolation of a Reactive Magnesium Phosphonoenolate, and Applications

By: Murata, Takatsugu; et al

Journal of Organic Chemistry (2024), 89(21), 15414-15435.

Scheme 47 (1 Reaction) Steps: **1** Yield: **98%**



Suppliers (59) Suppliers (95)

Double bond geometry shown

31-339-CAS-19339225 Steps: **1** Yield: **98%**

1.1 **Reagents:** Potassium carbonate
Catalysts: 1,8-Diazabicyclo[5.4.0]undec-7-ene
Solvents: Dimethylformamide, Dichloromethane; 24 - 36 h, 70 °C
1.2 **Solvents:** Water; 30 min, cooled

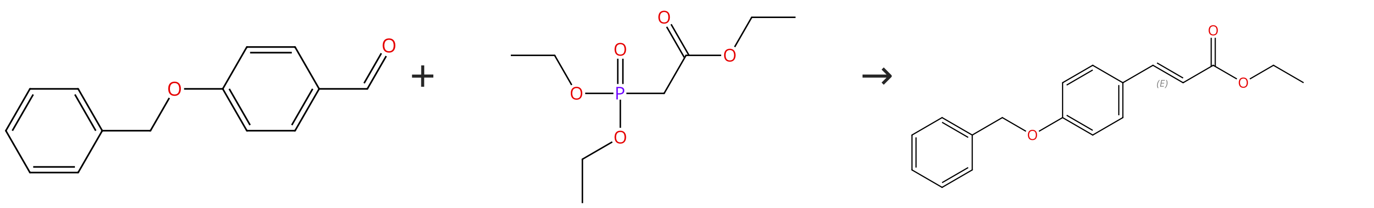
Experimental Protocols

Synthesis of cinnamic amide derivatives and their anti-melanogenic effect in α-MSH-stimulated B16F10 melanoma cells

By: Ullah, Sultan; et al

European Journal of Medicinal Chemistry (2019), 161, 78-92.

Scheme 48 (1 Reaction) Steps: **1** Yield: **98%**



Suppliers (92) Suppliers (95)

Double bond geometry shown

Suppliers (38)

31-339-CAS-19339223 Steps: **1** Yield: **98%**

1.1 **Reagents:** Potassium carbonate
Catalysts: 1,8-Diazabicyclo[5.4.0]undec-7-ene
Solvents: Dimethylformamide, Dichloromethane; 24 - 36 h, 70 °C
1.2 **Solvents:** Water; 30 min, cooled

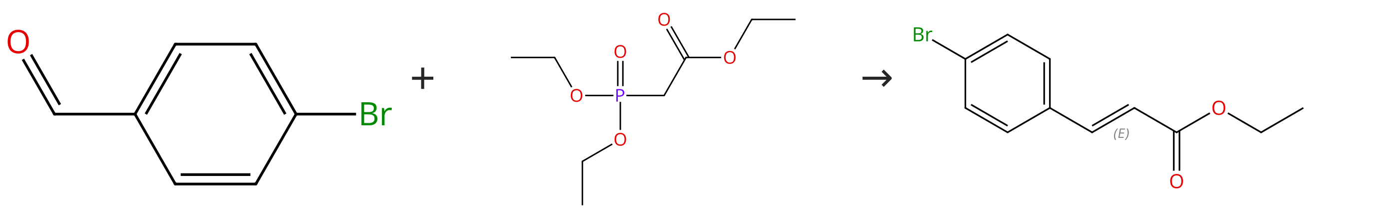
Experimental Protocols

Synthesis of cinnamic amide derivatives and their anti-melanogenic effect in α-MSH-stimulated B16F10 melanoma cells

By: Ullah, Sultan; et al

European Journal of Medicinal Chemistry (2019), 161, 78-92.

Scheme 49 (1 Reaction) Steps: **1** Yield: **98%**



Suppliers (91) Suppliers (95)

Double bond geometry shown

Suppliers (65)

31-614-CAS-41768726 Steps: **1** Yield: **98%**

1.1 **Reagents:** Potassium carbonate
Solvents: Ethanol; 20 min, 140 °C

Experimental Protocols

Microwave-Assisted Hydrolysis of Ethyl Azolylacetates and Cinnamates with K_2CO_3: Synthesis of Potassium Carboxylates

By: Hernandez-Fernandez, Jorge; et al

ACS Omega (2024), 9(39), 40783-40789.